



PH 421: Electromagnetic Theory

Course Code	PH 421
Course Name	Electromagnetic Theory
L-T-P-C	4-0-0-4
Prerequisites	Basic undergraduate physics courses and faculty consent
Intended for	I-Ph.D, M.Sc., B.Tech 3rd and 4th Year
Distribution	Core for I-Ph.D. and elective for others
Semester	Even
One-time approval	recommended by the 14th BoA (Sep 2016)

Course Contents

- **Electrostatics**
 - Differential equation for electric field, Poisson and Laplace equations, Formal solution for potential with Green's functions, Boundary value problems, Solutions of Laplace equation in cylindrical and spherical coordinates by orthogonal functions, Dielectrics, Polarization of a medium, Electrostatic energy. [9 Lectures]
- **Magnetostatics**
 - Biot-Savart law, Differential equation for static magnetic field, Vector potential, Magnetic field from localized current distributions, Examples of magnetostatic problems, Faraday's law of induction, Magnetic energy of steady current distributions. [7 Lectures]
- **Maxwell's Equations**
 - Displacement current, Maxwell's equations, Vector and Scalar potentials, Gauge symmetry, Coulomb and Lorentz gauges, Electromagnetic energy and momentum, Conservation laws. [5 Lectures]
- **Electromagnetic Waves**
 - Plane waves in a dielectric medium, Reflection and Refraction at dielectric interfaces, Frequency dispersion in dielectrics and metals, Dielectric constant and anomalous dispersion, Wave propagation in one dimension, Group velocity, Metallic wave guides, Boundary conditions at metallic surfaces, Propagation modes in wave guides, Resonant modes in cavities. [8 Lectures]
- **Electromagnetic Radiation**
 - Radiation, Electric dipole radiation, Magnetic dipole radiation, Radiation from a localized charge, The Lienard-Wiechert potentials, Radiation from an accelerated point charge. Abraham-Lorentz formula. [8 Lectures]
- **Relativistic Electrodynamics**
 - The special theory of relativity, Radius four-vector in contravariant and covariant form, Four-vector fields, Minkowski space, Covariant classical electrodynamics. [5 Lectures]

Text Books

1. J.D. Jackson, **Classical Electrodynamics**, John Wiley & Sons Inc., 1999.
2. D.J. Griffiths, **Introduction to Electrodynamics**, Prentice Hall, 1999.
3. W. Greiner, **Classical Electrodynamics**, Springer, 1998.
4. J.R. Reitz, F.J. Milford and R.W. Christy, **Foundations of Electromagnetic Theory**, Addition-Wesley, 2008.

References

1. L.D. Landau, E.M. Lifshitz and L.P. Pitaevskii, **Electrodynamics of Continuous Media**, Elsevier, 2010.
2. Feynman, Leighton, Sands, **The Feynman Lectures on Physics**, CALTECH, 2013.